

TABLE V
Reflow Ovens from Different Manufacturers

Manufacturer	Product
LPKF Laser and Electronics	ProtoFlow S
T-tech	SMT reflow oven -A
Mechatronics Systems	RK320, RK360
SMTnet	Manncorp 850 Benchtop, Manncorp BT300CP Benchtop
Essectec	RO-06-PLUS

To check the specs of all the three systems separately to see if they suit your requirements. Table IV shows some desktop pick-and-place machines from different manufacturers.

Reflow ovens and wave-soldering machines

The solder paste is applied and components are placed on the solder beds by the pick-and-place machines. But these components are not soldered to the



Fig. 5: A wave-soldering machine

pads as yet. For that you need to put these boards in an oven that heats up the boards and melts the solder paste. The melted solder paste sticks the pins of the components to the pads when it cools down.

Ovens also come in automatic and manual forms. In an automatic system, you decide the environment setting and put the board in the oven. The board ejects after the required time automatically. In a manual oven, you set the temperature and put the board in and take it out yourself after an estimated time period. Fig. 4 shows

a reflow oven and Table V lists some ovens from different manufacturers.

Though most people prefer to mount through-hole components manually but, for faster turnaround, a desktop wave-soldering machine can be used. These machines have melted solder in a flat bed, and the boards are moved over it slowly so the through-hole solder points catch some solder. The solder sticks the pins with the pads, making a permanent joint on cooling. Fig. 5 shows a desktop wave-soldering machine.

With all these equipment available, you can reduce the prototype-development time drastically. This, in turn, can reduce the time to market, increasing your product's chances of being a success. If you do not want to spend money on all these equipment in one go, select those that you really need to increase the efficiency of your product development cycle. ●

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